

13.3 Limits and Continuity

- Limits of two-variable continuous functions.
 - **HW 11, 15; ex.** $\lim_{(x,y) \rightarrow (2,0)} (xy^5 - x)$
- Limits at the boundary points of two-variable functions.
 - **HW 19, 23; ex.** $\lim_{(x,y) \rightarrow (2,0)} \frac{yx - y}{y(x^2 - 1)}$
- Nonexistence of limits using the Two-Path Test.
 - **HW 25, 29;**
 - ex.** $\lim_{(x,y) \rightarrow (0,0)} \frac{x + 2y}{x - 2y}$, use lines $x = 0$ and then $y = 0$.
 - ex.** $\lim_{(x,y) \rightarrow (0,0)} \frac{y^3 + x^3}{xy^2}$, use lines $x = y$ and then $x = -y$.

13.4 Partial Derivatives

- Given a functions $f(x, y)$, find f_x , f_y , f_{xx} , f_{yy} , f_{xy} , and f_{yx} .
 - **HW 7-15, 17, 23, 27, 29.**

13.5 The Chain Rule

- Use the Chain rule to find the derivatives of functions with one independent variable.
 - **HW 7-10.** (only two-variable functions)

13.6 Directional Derivatives and the Gradient

- Compute the gradient of two-variable functions.
 - **HW 11, 13.**
- Find a unit vector that gives the direction of the steepest ascent and descent of a function.
- Find a vector that gives a direction of no change of a function.
 - **HW 21, 25.**

13.8 Maximum and Minimum Problems

- Find the critical points of a function, and use the Second Derivative Test to determine (when possible) whether each critical point corresponds to a local maximum, minimum, or saddle point.
 - **HW 15, 21, 27.**